Karam Matar

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Software Engineer (B.Sc.) with hands-on experience in full-stack development, software security, CI/CD, deployment, and systems programming. Comfortable working across the software lifecycle, from feature development and testing to debugging, integration, deployment, and technical collaboration.

Professional Experience

Kravi Analytics AB — **Lund, Sweden**
Full-Stack Developer Intern — *Aug 2026 – Oct 2026*

- Independently designed and built a containerized application from scratch for reviewing and classifying satellite vessel detections using Python, FastAPI/Uvicorn, SQLAlchemy, Alembic, PostgreSQL/PostGIS, Jinja2, and Docker.
- Designed and implemented a dynamic batch workflow where users create review batches by geographic zone and date range. Active batches continuously receive newly available detections, then progress through a quality-review gate once the date range has ended and all detections have been classified, with confirmed labels stored for downstream machine-learning use.
- Built the data-ingestion and synchronization workflow, integrating API clients and S3-based TIFF imagery, implementing scheduled cron jobs to fetch new detections and update affected batches, and creating a backfill command for importing historical detection data across arbitrary date ranges.
- Built the project's deployment infrastructure, configuring dedicated SSH-based GitHub authentication, Linux server deployment, Nginx reverse proxying, and UFW firewall rules to make the application accessible over the internet.

Softhouse — **Karlskrona, Sweden**
Software Developer Intern — *Jan 2026 – May 2026*

- Shipped features end-to-end across React, Express, Prisma, and MongoDB for TeamTemp, an employee survey platform, and contributed to team decisions on the technology stack and technical approach in an 11-person agile team.
- Identified and patched security vulnerabilities (server-side validation, input sanitisation); wrote Cypress E2E tests and integrated them into the Bitbucket CI/CD pipeline.
- Led customer demos and an internal office presentation; served as the primary point of contact between developers, customer, and management.

Trafikverket — **Remote via BTH**
Software Developer Intern — *Jan 2025 – May 2025*

- Refactored existing PowerShell-based dependency-security tooling into JavaScript as part of a six-person team, supporting reusable vulnerability scanning across repositories.
- Contributed to packaging the refactored tooling as a reusable Azure DevOps task that generates SBOM files in JSON format.
- Contributed to integrating automated security checks into Azure DevOps workflows to support vulnerability detection during builds.

Blekinge Institute of Technology — **Karlskrona, Sweden**
Teaching Assistant – C++ — *Sep 2024 – Jan 2025*

- Mentored first-year students through lab supervision and 1-on-1 tutoring in pointers, memory management, and OOP.

Education

Blekinge Institute of Technology — **Karlskrona, Sweden**
B.Sc. Software Engineering — *Aug 2023 – Jun 2026*
Thesis with Iquest AB: ACID Bot – autonomous multi-agent AI workflow for bug-fix verification and PR creation. Key courses: Compiler Design, Operating Systems, Algorithms & Data Structures.

Newcastle University — **Newcastle upon Tyne, UK**
Exchange Semester – Computer Science — *Sep 2025 – Jan 2026*
Courses: Computer Vision & AI, Algorithm Design, Security Programming.

Projects

ACID Bot (JavaScript, Git, CI/CD, Multi-Agent Systems): Autonomous workflow that fetches bug tasks from ClickUp, fixes issues in external repos via controlled Git access, verifies correctness, and opens pull requests – with bounded retries per agent for cost and safety control.

Compiler from Scratch (C++): End-to-end compiler: lexer, parser, semantic analysis, CFG-based IR, three-address code generation, and a custom virtual machine.

ShowcaseSERL (Next.js, React, TypeScript, Tailwind CSS): Full-stack web app for showcasing student and researcher projects with dynamic listings, detail views, and a kiosk display mode.

Skills

Programming: C++, Java, Python, C#, JavaScript, TypeScript, SQL

Web & Frameworks: FastAPI, SQLAlchemy, React, React Native/Expo, Next.js, SvelteKit, Express.js, Node.js, Prisma

DevOps & Data: Git, Docker, Docker Compose, Linux, Nginx, Alembic, Azure DevOps, Bitbucket Pipelines, CI/CD, PostgreSQL, MySQL, MongoDB

Testing: Cypress (E2E), Jest, Bun Test

Spoken Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

Driver License: AM/B

Awards

1st Place: Hollytech Hackathon – won a capture-the-flag competition.